INCH-POUND

MIL-DTL-55302/66G
w/AMENDMENT 2
26 May 2020
SUPERSEDING
MIL-DTL-55302/66G
w/AMENDMENT 1
31 March 2011

DETAIL SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, SOCKET CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND INTERMEDIATE POSITIONS OF 14, 24, 44, 54 AND 26, 36, 56, 66 FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

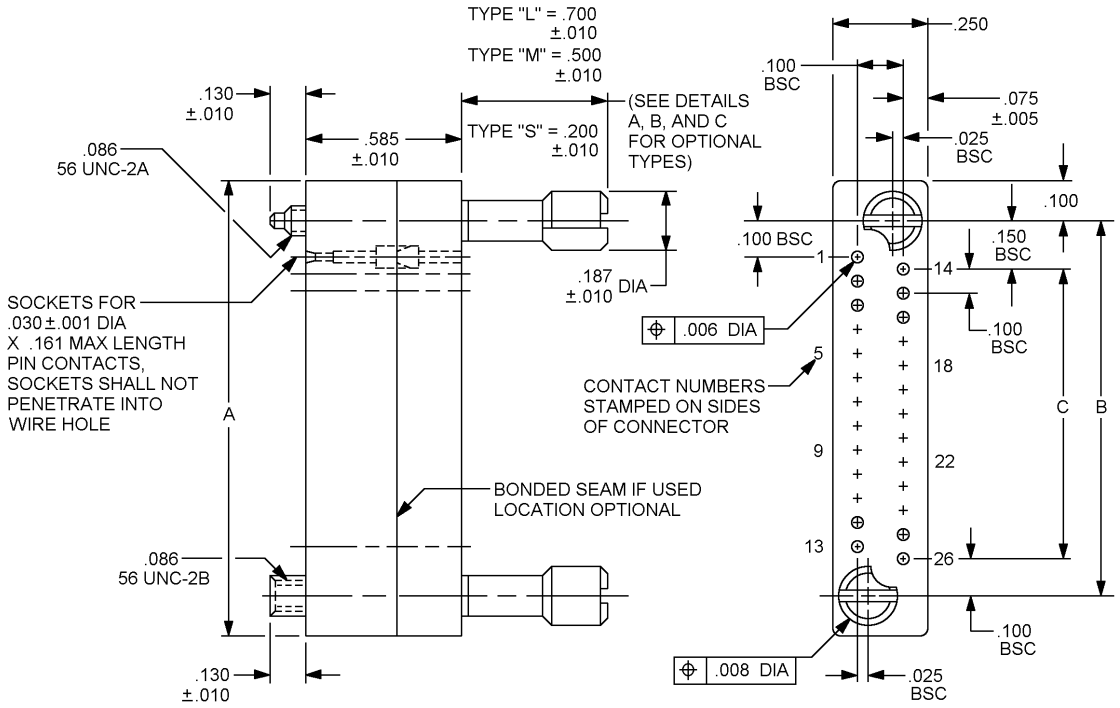
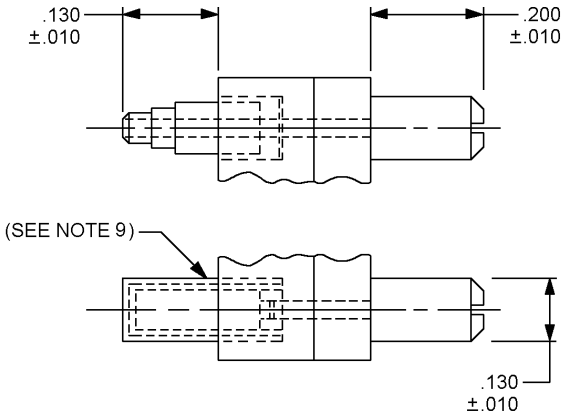



FIGURE 1. Connectors, receptacle, .100 (2.54 mm) spacing.

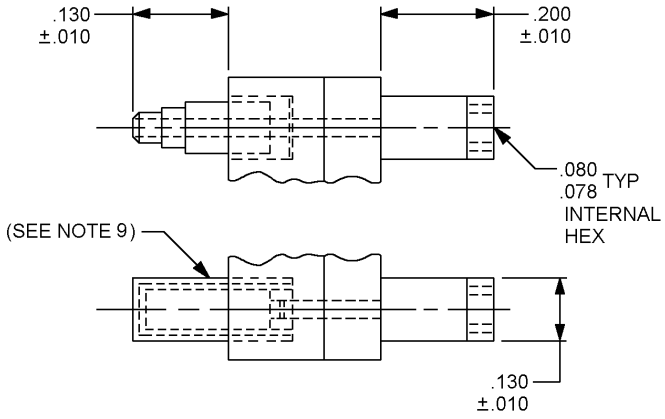
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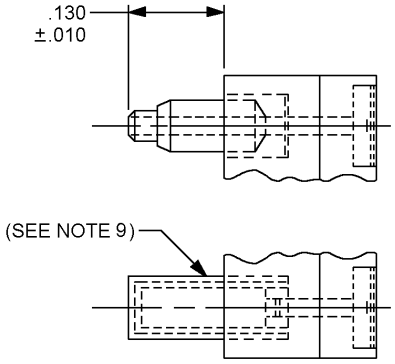
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TYPE "S" SHORT TURNING SLOTTED JACKSET
DETAIL A




TYPE "H" SHORT TURNING HEX JACKSET
DETAIL B




TYPE "F" FIXED JACKSET
DETAIL C

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .001 | 0.03 | .030 | 0.76 | .130 | 3.30 | .250 | 6.35 |
| .006 | 0.15 | .075 | 1.91 | .150 | 3.81 | .500 | 12.70 |
| .008 | 0.20 | .078 | 1.98 | .161 | 4.09 | .585 | 14.86 |
| .010 | 0.25 | .086 | 2.18 | .187 | 4.75 | .700 | 17.78 |
| .025 | 0.64 | .100 | 2.54 | .200 | 5.08 | | |

FIGURE 1. Connectors, receptacle, .100 (2.54 mm) spacing - Continued.



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| Number of contacts | A ±.010 | BSC B | REF C |
|---|---|---|---|
| 10 | .850 (21.59) | .650 (16.51) | .400 (10.16) |
| 14 | 1.050 (26.67) | .850 (21.59) | .600 (15.24) |
| 20 | 1.350 (34.29) | 1.150 (29.21) | .900 (22.86) |
| 24 | 1.550 (39.37) | 1.350 (34.29) | 1.100 (27.94) |
| 26 | 1.650 (41.91) | 1.450 (36.83) | 1.200 (30.48) |
| 30 | 1.850 (46.99) | 1.650 (41.91) | 1.400 (35.56) |
| 36 | 2.150 (54.61) | 1.950 (49.53) | 1.700 (43.18) |
| 40 | 2.350 (56.69) | 2.150 (54.61) | 1.900 (48.26) |
| 44 | 2.550 (64.77) | 2.350 (56.69) | 2.100 (53.34) |
| 50 | 2.850 (72.39) | 2.650 (67.31) | 2.400 (60.96) |
| 54 | 3.050 (77.47) | 2.850 (72.39) | 2.600 (66.04) |
| 56 | 3.150 (80.01) | 2.950 (74.93) | 2.700 (68.58) |
| 60 | 3.350 (85.09) | 3.150 (80.01) | 2.900 (73.66) |
| 66 | 3.650 (92.71) | 3.450 (87.63) | 3.200 (81.28) |
| 70 | 3.850 (97.79) | 3.650 (92.71) | 3.400 (86.36) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-DTL-55302/57, MIL-DTL-55302/61, and MIL-DTL-55302/63 using appropriate hardware.
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector. As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked.
6. Termination layout on .025 (0.64 mm) modular grid.
7. Metric equivalents are in parentheses.
8. Contacts M55302/65-01 (standard force) and M55302/65-02 (low force) are individual contacts shipped loose in a plastic bag unless otherwise specified (see MIL-DTL-55302/65). Contacts may be ordered separately.
9. Hole depth to female guide hardware is .282 min (7.16 mm). Full thread depth to female threaded hardware is .240 min (6.1 mm).

FIGURE 1. Connectors, receptacle, .100 (2.54 mm) spacing - Continued.



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Requirements:

Dimensions and configuration: See figure 1.

Material: Jackscrews and jack sockets shall be made of corrosion resisting stainless steel in accordance with ASTM A581/A581M or ASTM A582/A582M passivated in accordance with SAE-AMS2700 or shall be corrosion resistant copper nickel alloy (61 ±2% nickel, 2.5% max. other, and the balance zinc and shall have a brinell hardness of 145 to 175.

Plating: The contact plating shall be in accordance with MIL-DTL-55302 or gold in accordance with MIL-DTL-45204 or equivalent as approved by the preparing activities, type II, grade C, class 1, over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold in accordance with MIL-DTL-45204 or equivalent as approved by the preparing activities, type II, grade C, class 00, or type III, grade A, class 00 over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 50 to 150 microinches in the termination area.

Contact identification: See figure 1.

Contact engagement and separation forces (number 22 contacts):
Minimum separation: 0.5 ounce.
Maximum engagement: 12.0 ounces for standard force and 4.0 ounces for low force.

Mating and unmating:
Standard insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.56 and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Low insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.25 and the withdrawal force, in pounds, shall be a minimum of 0.04 times the number of contacts and shall not exceed the maximum insertion force.

NOTE: Not applicable when mating and unmating is accomplished through the engaging of turning jacksets.

Jackscrew and jack socket torque: 2-inch pounds maximum fully mated.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:
Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Current rating: 5.0 amperes maximum in accordance with contact.

Insertion and removal tool: SAE-AS81969/12.

Crimp tool: M22520/2-01



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Example of Part or Identifying Number (PIN):

| M55302/66 | L | 10 | L |
|---|---|---|---|
| Basic number of specification sheet | Type of contacts | Number of contacts | Type of mounting hardware (see figure 1) |
| | L = Low insertion force contacts<br>- = Standard insertion force contacts | (Contacts supplied with connector are crimp removable type for no. 22 maximum wire size .048 maximum insulation) | L = .700 slotted turning jackset<br>M = .500 slotted turning jackset<br>S = .200 slotted turning jackset<br>F = fixed jackset<br>H = hex turning jackset |

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following documents:

MIL-DTL-45204
MIL-DTL-55302/57
MIL-DTL-55302/61
MIL-DTL-55302/63
MIL-DTL-55302/65
SAE-AS81969/12
ASTM A581/A581M
ASTM A582/A582M
SAE-AMS2700
SAE-AMS-QQ-N-290

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2020-087)

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, MC, OS
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil/.



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